1) The value of V_o is:

a) 7.4 V b) 574 mV

c) 500 mV d) -6400 mV

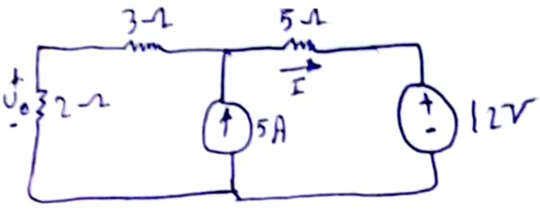

Fig. 1

2) The value of the current (i) in 5Ω is:

a) -0.75 A b) 1.4 A

c) 1.3 A d) 0.7 A

3) The power absorbed by 3Ω resister is:

a) 5.23 W b) 6.46 W

c) 10.46 W d) 41.07 W

4) The Power absorbed by 12V voltage source is:

a) 31.2 W b) 44.8 W

c) -15.6 W d) -44.8 W

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Prob. 2: For question 5,6 consider the current through a 1 mH inductor is $i(t) = 60 \cos 100t$ mA.

5) The terminal voltage across the inductor is:

a) $-6 \sin 100t$ mV b) $-6 \cos 100t$ mV

c) -9V d) 6 mV

6) The Energy stored in the inductor is:

a) $18 \sin^2(100)$ μJ b) $1.8 \cos^2(100t)$ μJ

c) 180 μJ d) 18 mJ

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Prob 3: For question 7, 8, 9, 10 Consider the circuit shown in Figure 2. The switch has been closed for a long time. At $t=0$, the switch is opene

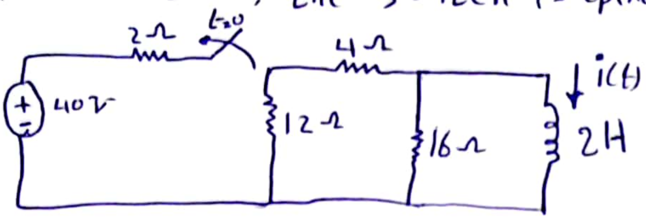

Fig.2

7) The initial current $i(t=0+)$ is:

a) 24 A b) 60 A

c) 36 A d) 6 A

8) The time constant of the circuit is:

a) 120 s b) 250 ms

c) 60 s d) 500 ms

9) The current in the inductor at $t=500$ ms is:

a) 128.8 mA b) 0.812 A

c) 606.5 mA d) 2.8 A

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10) The final value of the voltage $V(t=\infty)$ across the inductor is:

a) 0V b) 36V

c) 24V d) 12V

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Prob. 4: Consider the circuit ... figure 3. use the Thevenin's theorem with respect to terminals (a-b)

a. The Thevenin voltage is ______

b. The Thevenin resistance is ______

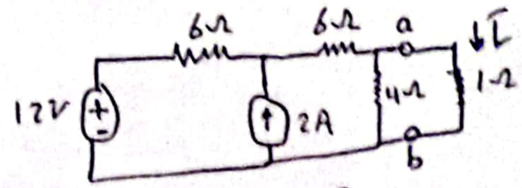


Fig 3

c. The current I is ______

d. what is the value of the load resistance that should be connected to the ab terminal to achieve maximum power transfer? ______

Prob. 5: Consider the circuit shown in figure 4. And answer

a- The value of α (damping factor) is:

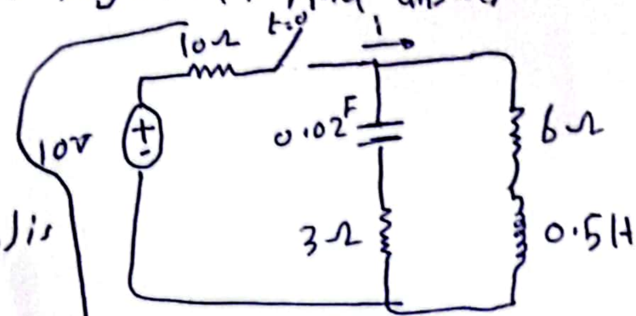


Fig. 4

b- The value of w_0 (resonant frequency) is

c. The value of s_1 is ______

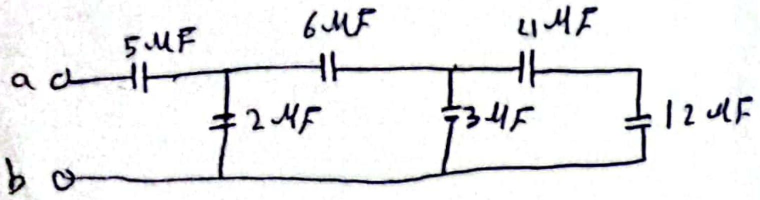
d. The value of s_2 is ______

e- The natural response $i(t)$ for $t>0$ in general form is ______

f- The initial inductor current $i(0)$ is ______

g- The initial capacitor voltage $v(0)$ is ______

Prob 6: a) Refer to figure 5; the Value of the equivalent capacitance is: ______



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b) Refer to figure 6; the value of the equivalent resistance is:

____

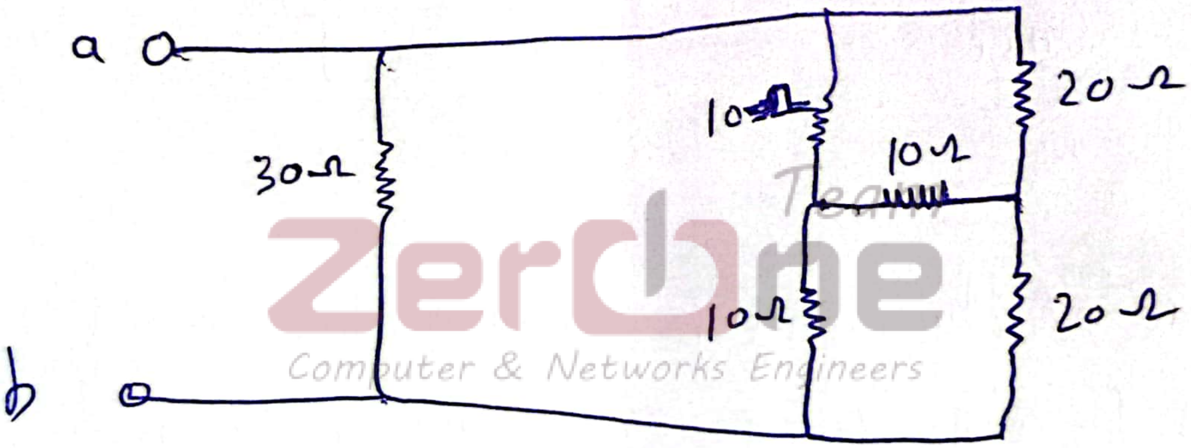

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Fig. 6

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